PREDICTIVE INSIGHT ENGINE · SUPERVISED LEARNING PROJECT

Part A: Conceptual Understanding

Theory answers on Supervised Learning, Regression, Linear Regression Assumptions, Bias-Variance Trade-Off, Overfitting and Underfitting

1

What are Supervised Learning Algorithms?

Supervised learning algorithms learn from **labeled data**, where every input (features) comes with the correct output (target). The model learns the pattern that links inputs to outputs, and then uses it to **predict the output for new, unseen data**. It is called *supervised* because the known answers guide the learning and the model's errors are measured and corrected.

Examples: Linear Regression, Logistic Regression, Decision Trees, KNN, Support Vector Machines.

> **In this project:** features such as area, bedrooms and location score (input) are used to predict the house price (output).

2

Difference between Regression Algorithms vs Classification Algorithms

Regression predicts a **continuous numeric value**, while **classification** predicts a **discrete category** (class label). The choice depends on the type of the target variable.

Point	Regression	Classification
Output	Continuous number	Category / class label
Example	House price in INR, temperature	Spam or not spam, disease or no disease
Common algorithms	Linear, Polynomial, Ridge, Lasso	Logistic Regression, Decision Tree, KNN, SVM
Evaluation metrics	MSE, MAE, RMSE, R^2	Accuracy, Precision, Recall, F1-score
This project	**Regression** (target: house price)	Not used

3 Explain Simple Linear Regression

Simple Linear Regression models the relationship between **one independent variable (x)** and **one dependent variable (y)** using a straight line:

> $y = b_0 + b_1 x + e$ where b_0 = intercept (value of y when x = 0), b_1 = slope (change in y for every 1-unit increase in x), **e** = error term.

The best line is found with the **Least Squares Method**, which minimizes the sum of squared differences between actual and predicted values. **Example from this project:** Price = -11,63,519 + 14,788 x Area, so every extra 1 sq.ft adds about Rs. 14,788 to the price.

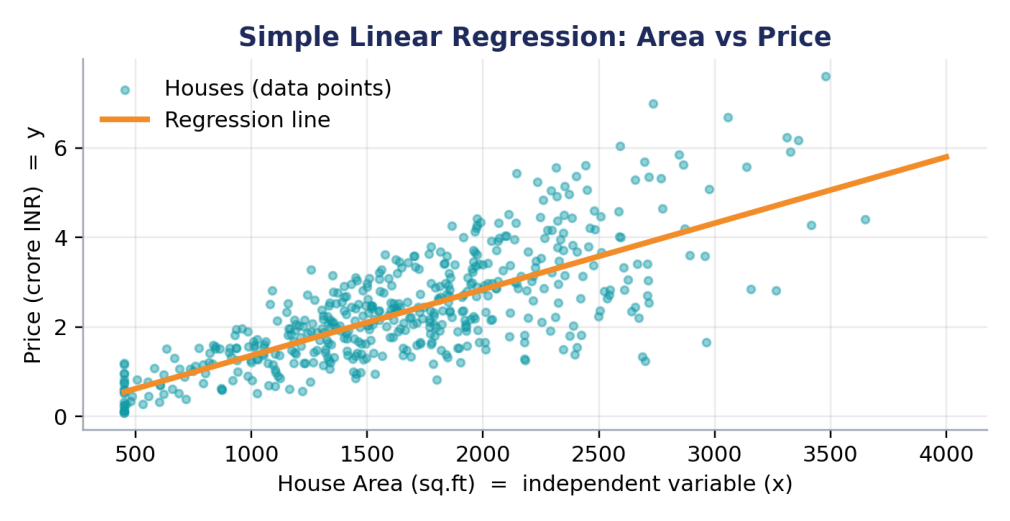


Figure: Regression line fitted on the project dataset (area vs house price).

4 List and explain the Assumptions of Linear Regression

Linear Regression gives reliable results only when these **five assumptions** hold:

Assumption	Explanation	How to check
1. Linearity	The relationship between the features and the target is linear (a straight-line trend).	Scatter plot, Actual vs Predicted plot
2. Independence of errors	Residuals are independent of each other, with no pattern or autocorrelation.	Durbin-Watson test (value near 2)
3. Homoscedasticity	Residuals have a constant spread (equal variance) at every level of the prediction.	Residuals vs Predicted plot, Breusch-Pagan test
4. Normality of errors	Residuals follow an approximately normal (bell-shaped) distribution.	Histogram, Q-Q plot
5. No multicollinearity	Independent variables are not highly correlated with each other (applies to multiple regression).	Correlation heatmap, VIF (below 5 to 10)

5 What is Bias-Variance Trade-Off?

It is the balance between two sources of model error. **Bias** is the error from over-simple assumptions (the model misses the real pattern, causing **underfitting**). **Variance** is the error from being too sensitive to the training data (the model changes a lot with new data, causing **overfitting**).

A more complex model **lowers bias but raises variance**, so the goal is to find the point where the total error ($Bias^2$ + Variance + Noise) is the lowest.

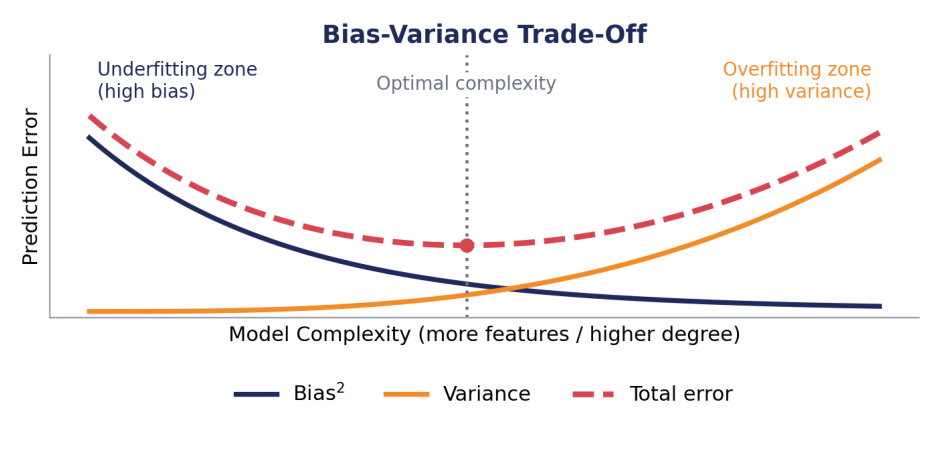


Figure: Total error is lowest at the right model complexity, between underfitting and overfitting.

6 Explain Overfitting and Underfitting with examples

Underfitting happens when the model is too simple to capture the pattern, so it performs poorly on **both** training and test data. **Overfitting** happens when the model is too complex and memorizes the training data including its noise, so it performs very well on training data but poorly on **new** data.



Figure: The same data fitted with a straight line (underfit), a smooth curve (good fit) and a very wiggly curve (overfit).

	Underfitting	Overfitting
Cause	Model too simple, too few features	Model too complex, too many features or terms
Train vs Test	Both scores are low	Train score high, test score much lower
Bias / Variance	High bias, low variance	Low bias, high variance
Example (this project)	Simple Linear Regression using only area: Train R^2 = 0.572, Test R^2 = 0.563	Polynomial Regression of degree 4 (1000 features): Train R^2 = 0.978 but Test R^2 = 0.927
How to fix	Add more features, use a more flexible model	Use a simpler model, regularization, more data, cross-validation